4	**(a)**	Interference is the <u>superposing of two or more waves</u> to produce <u>regions of maxima and minima in space</u>, according to the principle of superposition.	B1
	(b)(i)	Using dsinθ = nλ, let θ = 90° and choose the largest λ in order for a full spectrum will be observed, $\frac{1}{3\times10^{5}}\sin 90 = n(700\times10^{-9})$ $n = 4.8$ Hence, the maximum order for a full spectrum to be observed is 4.	M1 A1
	(b)(ii)	Considering the first order diffraction angle for 700 nm wavelength, $\frac{1}{3\times10^{5}}\sin\theta = 1(700\times10^{-9})$ $\theta = 12.1^{o}$ Considering the second order diffraction angle for 400 nm wavelength, $\frac{1}{3\times10^{5}}\sin\theta = 2(400\times10^{-9})$ $\theta = 13.9^{o}$ Since the second order diffraction angle for 400 nm wavelength is larger than the first order diffraction angle for 700 nm wavelength, the spectra do not overlap.	C1 C1 A1

	(c)	White light consists of visible light of different wavelengths. There is no diffraction of light at the central region so this region remains white. In diffraction of waves, waves with larger wavelength will have larger diffracted angles. Light sources with lower wavelength do not diffract as much and do not achieve overlapping of light sources at the edge to produce white light.	B1 B1

Qn		**Answer**	**Mark**
5	**(c)(i)**	Resistance of copper, R	